MASTER MAD — ANDROID JAVA + XML EXAM PATTERN

Reusable-pattern program for Mobile Application Development Lab

Goal: Learn one reusable master structure and then change the field names, arrays, messages and Intent destination according to the exam question.

Technology: Java + XML. Beginner-friendly Android programming. No Kotlin, Compose or SQLite.

Memorization flow: Form → Validation → Widgets → Preferences → Dialog → Bundle → Explicit Intent → Second Activity → ListView → Adapter → Click → AlertDialog/Toast → Long Click → Context Menu → Implicit Intent → Fragment → Bundle → Dynamic UI.

1. WHAT YOU MUST MASTER

LEVEL 1 — MUST KNOW PERFECTLY

Login, Calculator, Explicit Intent, Activity Lifecycle, RadioButton + CheckBox, Implicit Intent all actions, ListView + ArrayAdapter, ScrollView + dynamic addView, Spinner + Adapter + OnItemSelectedListener, SharedPreferences.

LEVEL 2 — VERY IMPORTANT

AlertDialog, PopupMenu, Options Menu, Context Menu, Custom Dialog, Long-click, notifyDataSetChanged, TextWatcher, RatingBar, DatePickerDialog, TimePickerDialog, IME Done, Fragments, FragmentManager, replace, addToBackStack, setArguments, getArguments, Bundle, dynamic UI, ProgressBar, AutoCompleteTextView, TableLayout, GridLayout.

LEVEL 3 — EXAM MASTER PATTERN

Activity1 → Form widgets → Validation → Options Menu/Button → AlertDialog → SharedPreferences → Bundle → Explicit Intent → Activity2 → Read preferences/Intent data → ListView + Adapter → Normal click → AlertDialog/Toast → Long press → Context Menu → Implicit Intent Browser/Dialer/Email/Maps/Calendar.

2. PROJECT STRUCTURE

```
MasterMAD/
├── app/
│   └── src/main/
│       ├── java/com/example/mastermad/
│       │   ├── MainActivity.java
│       │   ├── DashboardActivity.java
│       │   ├── HomeFragment.java
│       │   └── DetailsFragment.java
│       ├── res/
│       │   ├── layout/
│       │   │   ├── activity_main.xml
│       │   │   ├── activity_dashboard.xml
│       │   │   ├── fragment_home.xml
│       │   │   ├── fragment_details.xml
│       │   │   └── dialog_add_item.xml
│       │   └── menu/
│       │       ├── menu_main.xml
│       │       └── menu_dashboard.xml
│       └── AndroidManifest.xml
```

3. MASTER JAVA — ACTIVITY 1

This activity combines the main reusable patterns: form widgets, validation, SharedPreferences, Spinner, AutoCompleteTextView, TextWatcher, IME Done, date/time dialogs, RatingBar, calculator, dynamic addView, Options Menu, AlertDialog, custom dialog, Bundle, Explicit Intent and lifecycle.

```
package com.example.mastermad;

import android.app.*;
import android.content.*;
import android.os.Bundle;
import android.text.*;
import android.text.method.PasswordTransformationMethod;
import android.util.Log;
import android.view.*;
import android.view.inputmethod.EditorInfo;
import android.widget.*;
import java.text.SimpleDateFormat;
import java.util.*;

public class MainActivity extends Activity {

    private EditText nameEditText, emailEditText, passwordEditText;
    private EditText numberOneEditText, numberTwoEditText, commentEditText;
    private TextView resultTextView, dateTextView, timeTextView, dynamicTextView;
    private RadioGroup genderRadioGroup;
    private CheckBox readingCheckBox, sportsCheckBox;
    private Spinner courseSpinner;
    private AutoCompleteTextView cityAutoCompleteTextView;
    private RatingBar ratingBar;
    private LinearLayout dynamicContainer;

    private SharedPreferences sharedPreferences;
```

```
private final String[] courses = {
        "MCA", "MBA", "MSc Computer Science", "BCA"
};

private final String[] cities = {
        "Thiruvananthapuram", "Kochi", "Kollam",
        "Kozhikode", "Kannur"
};

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    // Form widgets
    nameEditText = findViewById(R.id.nameEditText);
    emailEditText = findViewById(R.id.emailEditText);
    passwordEditText = findViewById(R.id.passwordEditText);
    numberOneEditText = findViewById(R.id.numberOneEditText);
    numberTwoEditText = findViewById(R.id.numberTwoEditText);
    commentEditText = findViewById(R.id.commentEditText);

    resultTextView = findViewById(R.id.resultTextView);
    dateTextView = findViewById(R.id.dateTextView);
    timeTextView = findViewById(R.id.timeTextView);
    dynamicTextView = findViewById(R.id.dynamicTextView);

    genderRadioGroup = findViewById(R.id.genderRadioGroup);
    readingCheckBox = findViewById(R.id.readingCheckBox);
    sportsCheckBox = findViewById(R.id.sportsCheckBox);

    courseSpinner = findViewById(R.id.courseSpinner);
    cityAutoCompleteTextView = findViewById(R.id.cityAutoCompleteTextView);
    ratingBar = findViewById(R.id.ratingBar);
    dynamicContainer = findViewById(R.id.dynamicContainer);

    sharedPreferences = getSharedPreferences("MasterPrefs", MODE_PRIVATE);

    setupSpinner();
    setupAutoComplete();
    setupTextWatcher();
    setupImeDone();

    findViewById(R.id.saveButton).setOnClickListener(v -> saveUser());
    findViewById(R.id.showSavedButton).setOnClickListener(v -> showSavedUser());
    findViewById(R.id.loginButton).setOnClickListener(v -> openDashboard());

    findViewById(R.id.addDynamicButton).setOnClickListener(v -> addDynamicView());

    findViewById(R.id.addButton).setOnClickListener(v -> calculate("+"));
    findViewById(R.id.subtractButton).setOnClickListener(v -> calculate("-"));
    findViewById(R.id.multiplyButton).setOnClickListener(v -> calculate("*"));
    findViewById(R.id.divideButton).setOnClickListener(v -> calculate("/"));
    findViewById(R.id.clearButton).setOnClickListener(v -> resultTextView.setText(""));

    findViewById(R.id.dateButton).setOnClickListener(v -> showDatePicker());
    findViewById(R.id.timeButton).setOnClickListener(v -> showTimePicker());

    findViewById(R.id.alertButton).setOnClickListener(v -> showAlertDialog());
    findViewById(R.id.customDialogButton).setOnClickListener(v -> showCustomDialog());
}

// ---------------- VALIDATION + EXPLICIT INTENT ----------------

private boolean validateForm() {
    String name = nameEditText.getText().toString().trim();
    String email = emailEditText.getText().toString().trim();
    String password = passwordEditText.getText().toString().trim();

    if (name.isEmpty()) {
        nameEditText.setError("Enter name");
        nameEditText.requestFocus();
        return false;
    }

    if (email.isEmpty()) {
        emailEditText.setError("Enter email");
        emailEditText.requestFocus();
        return false;
    }

    if (password.isEmpty()) {
        passwordEditText.setError("Enter password");
        passwordEditText.requestFocus();
        return false;
    }

    if (password.length() < 6) {
        passwordEditText.setError("Minimum 6 characters");
        passwordEditText.requestFocus();
        return false;
```

```
    }

    return true;
}

private void openDashboard() {
    if (!validateForm()) return;

    String name = nameEditText.getText().toString().trim();
    String email = emailEditText.getText().toString().trim();

    Intent intent = new Intent(MainActivity.this, DashboardActivity.class);

    Bundle bundle = new Bundle();
    bundle.putString("name", name);
    bundle.putString("email", email);
    intent.putExtras(bundle);

    startActivity(intent);
}

// ---------------- SHARED PREFERENCES ----------------

private void saveUser() {
    if (!validateForm()) return;

    String name = nameEditText.getText().toString().trim();
    String email = emailEditText.getText().toString().trim();

    SharedPreferences.Editor editor = sharedPreferences.edit();
    editor.putString("name", name);
    editor.putString("email", email);
    editor.apply();

    Toast.makeText(this, "Data saved", Toast.LENGTH_SHORT).show();
}

private void showSavedUser() {
    String name = sharedPreferences.getString("name", "No name");
    String email = sharedPreferences.getString("email", "No email");

    new AlertDialog.Builder(this)
            .setTitle("Saved Details")
            .setMessage("Name: " + name + "\nEmail: " + email)
            .setPositiveButton("OK", null)
            .show();
}

// ---------------- SPINNER ----------------

private void setupSpinner() {
    ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_spinner_item,
            courses
    );

    adapter.setDropDownViewResource(
            android.R.layout.simple_spinner_dropdown_item
    );

    courseSpinner.setAdapter(adapter);

    courseSpinner.setOnItemSelectedListener(
            new AdapterView.OnItemSelectedListener() {
                @Override
                public void onItemSelected(
                        AdapterView<?> parent, View view,
                        int position, long id) {

                    Toast.makeText(
                            MainActivity.this,
                            "Selected: " + courses[position],
                            Toast.LENGTH_SHORT
                    ).show();
                }

                @Override
                public void onNothingSelected(AdapterView<?> parent) {
                }
            }
    );
}

// ---------------- AUTOCOMPLETE ----------------

private void setupAutoComplete() {
    ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_dropdown_item_1line,
            cities
```

```
    );

    cityAutoCompleteTextView.setAdapter(adapter);
    cityAutoCompleteTextView.setThreshold(1);
}

// ---------------- TEXTWATCHER ----------------

private void setupTextWatcher() {
    commentEditText.addTextChangedListener(
            new TextWatcher() {
                @Override
                public void beforeTextChanged(
                        CharSequence s, int start,
                        int count, int after) {
                }

                @Override
                public void onTextChanged(
                        CharSequence s, int start,
                        int before, int count) {

                    dynamicTextView.setText(
                            "Characters: " + s.length()
                    );
                }

                @Override
                public void afterTextChanged(Editable s) {
                }
            }
    );
}

// ---------------- IME DONE ----------------

private void setupImeDone() {
    commentEditText.setOnEditorActionListener(
            (v, actionId, event) -> {
                if (actionId == EditorInfo.IME_ACTION_DONE) {
                    v.clearFocus();
                    return true;
                }
                return false;
            }
    );
}

// ---------------- DATE PICKER ----------------

private void showDatePicker() {
    Calendar calendar = Calendar.getInstance();

    DatePickerDialog dialog = new DatePickerDialog(
            this,
            (view, year, month, dayOfMonth) -> {
                String date = dayOfMonth + "/" +
                        (month + 1) + "/" + year;
                dateTextView.setText("Date: " + date);
            },
            calendar.get(Calendar.YEAR),
            calendar.get(Calendar.MONTH),
            calendar.get(Calendar.DAY_OF_MONTH)
    );

    dialog.getDatePicker().setMinDate(
            System.currentTimeMillis()
    );

    dialog.show();
}

// ---------------- TIME PICKER ----------------

private void showTimePicker() {
    Calendar calendar = Calendar.getInstance();

    TimePickerDialog dialog = new TimePickerDialog(
            this,
            (view, hourOfDay, minute) -> {
                String time = String.format(
                        Locale.getDefault(),
                        "%02d:%02d",
                        hourOfDay, minute
                );
                timeTextView.setText("Time: " + time);
            },
            calendar.get(Calendar.HOUR_OF_DAY),
            calendar.get(Calendar.MINUTE),
            true
    );
```

```
    dialog.show();
}

// ---------------- RATING BAR ----------------

private float getRating() {
    return ratingBar.getRating();
}

// ---------------- CALCULATOR ----------------

private void calculate(String operator) {
    try {
        double first = Double.parseDouble(
                numberOneEditText.getText().toString().trim()
        );

        double second = Double.parseDouble(
                numberTwoEditText.getText().toString().trim()
        );

        double answer;

        switch (operator) {
            case "+":
                answer = first + second;
                break;

            case "-":
                answer = first - second;
                break;

            case "*":
                answer = first * second;
                break;

            case "/":
                if (second == 0) {
                    resultTextView.setText("Cannot divide by zero");
                    return;
                }
                answer = first / second;
                break;

            default:
                return;
        }

        resultTextView.setText(String.valueOf(answer));

    } catch (NumberFormatException e) {
        resultTextView.setText("Enter valid numbers");
    }
}

// ---------------- DYNAMIC UI ----------------

private void addDynamicView() {
    TextView textView = new TextView(this);

    textView.setText("Dynamic item " +
            (dynamicContainer.getChildCount() + 1));

    textView.setTextSize(16);
    textView.setPadding(12, 12, 12, 12);

    dynamicContainer.addView(textView);
}

// ---------------- ALERT DIALOG ----------------

private void showAlertDialog() {
    new AlertDialog.Builder(this)
            .setTitle("Confirmation")
            .setMessage("Do you want to continue?")
            .setPositiveButton("Yes", (dialog, which) ->
                    Toast.makeText(
                            this,
                            "Yes selected",
                            Toast.LENGTH_SHORT
                    ).show()
            )
            .setNegativeButton("No", null)
            .show();
}

// ---------------- CUSTOM DIALOG ----------------

private void showCustomDialog() {
    View view = getLayoutInflater().inflate(
            R.layout.dialog_add_item, null
```

```
    );

    EditText itemNameEditText =
            view.findViewById(R.id.itemNameEditText);

    EditText quantityEditText =
            view.findViewById(R.id.quantityEditText);

    new AlertDialog.Builder(this)
            .setTitle("Add Item")
            .setView(view)
            .setPositiveButton("Save", (dialog, which) -> {

                String itemName =
                        itemNameEditText.getText().toString().trim();

                String quantity =
                        quantityEditText.getText().toString().trim();

                if (itemName.isEmpty() ||
                        quantity.isEmpty()) {

                    Toast.makeText(
                            this,
                            "Enter all details",
                            Toast.LENGTH_SHORT
                    ).show();

                } else {
                    Toast.makeText(
                            this,
                            "Item added",
                            Toast.LENGTH_SHORT
                    ).show();
                }
            })
            .setNegativeButton("Cancel", null)
            .show();
}

// ---------------- OPTIONS MENU ----------------

@Override
public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(
            R.menu.menu_main,
            menu
    );
    return true;
}

@Override
public boolean onOptionsItemSelected(MenuItem item) {

    if (item.getItemId() == R.id.menu_save) {
        saveUser();
        return true;
    }

    if (item.getItemId() == R.id.menu_about) {
        showAlertDialog();
        return true;
    }

    return super.onOptionsItemSelected(item);
}

// ---------------- ACTIVITY LIFECYCLE ----------------

@Override
protected void onStart() {
    super.onStart();
    Log.d("LIFECYCLE", "onStart");
}

@Override
protected void onResume() {
    super.onResume();
    Log.d("LIFECYCLE", "onResume");
}

@Override
protected void onPause() {
    super.onPause();
    Log.d("LIFECYCLE", "onPause");
}

@Override
protected void onStop() {
    super.onStop();
    Log.d("LIFECYCLE", "onStop");
```

```
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d("LIFECYCLE", "onRestart");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d("LIFECYCLE", "onDestroy");
    }
}
```

4. MASTER XML — ACTIVITY 1

The layout deliberately demonstrates ScrollView, LinearLayout, TableLayout, GridLayout, EditText, RadioGroup/RadioButton, CheckBox, Spinner, AutoCompleteTextView, RatingBar, buttons and a dynamic container.

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Master MAD Form"
            android:textSize="24sp"
            android:textStyle="bold" />

        <TableLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:stretchColumns="1">

            <TableRow>
                <TextView
                    android:text="Name"
                    android:padding="8dp" />

                <EditText
                    android:id="@+id/nameEditText"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:hint="Enter name"
                    android:inputType="textPersonName" />
            </TableRow>

            <TableRow>
                <TextView
                    android:text="Email"
                    android:padding="8dp" />

                <EditText
                    android:id="@+id/emailEditText"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:hint="Enter email"
                    android:inputType="textEmailAddress" />
            </TableRow>

            <TableRow>
                <TextView
                    android:text="Password"
                    android:padding="8dp" />

                <EditText
                    android:id="@+id/passwordEditText"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:hint="Enter password"
                    android:inputType="textPassword" />
            </TableRow>

        </TableLayout>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Gender" />

        <RadioGroup
            android:id="@+id/genderRadioGroup"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">

            <RadioButton
                android:id="@+id/maleRadioButton"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Male" />

            <RadioButton
```

```
        android:id="@+id/femaleRadioButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female" />
</RadioGroup>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Hobbies" />

<CheckBox
    android:id="@+id/readingCheckBox"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Reading" />

<CheckBox
    android:id="@+id/sportsCheckBox"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Sports" />

<Spinner
    android:id="@+id/courseSpinner"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

<AutoCompleteTextView
    android:id="@+id/cityAutoCompleteTextView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="City"
    android:inputType="text" />

<RatingBar
    android:id="@+id/ratingBar"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:numStars="5"
    android:stepSize="1" />

<EditText
    android:id="@+id/commentEditText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Comment"
    android:inputType="textMultiLine"
    android:imeOptions="actionDone"
    android:minLines="3" />

<TextView
    android:id="@+id/dynamicTextView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Characters: 0" />

<GridLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:columnCount="2">

    <Button
        android:id="@+id/saveButton"
        android:text="Save" />

    <Button
        android:id="@+id/showSavedButton"
        android:text="Show Saved" />

    <Button
        android:id="@+id/loginButton"
        android:text="Open Dashboard" />

    <Button
        android:id="@+id/alertButton"
        android:text="AlertDialog" />

    <Button
        android:id="@+id/customDialogButton"
        android:text="Custom Dialog" />

    <Button
        android:id="@+id/addDynamicButton"
        android:text="Add Dynamic View" />
</GridLayout>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Date and Time" />
```

```
        <Button
            android:id="@+id/dateButton"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Select Date" />

        <TextView
            android:id="@+id/dateTextView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content" />

        <Button
            android:id="@+id/timeButton"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Select Time" />

        <TextView
            android:id="@+id/timeTextView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Calculator"
            android:textStyle="bold" />

        <EditText
            android:id="@+id/numberOneEditText"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="First number"
            android:inputType="numberDecimal" />

        <EditText
            android:id="@+id/numberTwoEditText"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Second number"
            android:inputType="numberDecimal" />

        <GridLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:columnCount="4">

            <Button
                android:id="@+id/addButton"
                android:text="+" />

            <Button
                android:id="@+id/subtractButton"
                android:text="-" />

            <Button
                android:id="@+id/multiplyButton"
                android:text="*" />

            <Button
                android:id="@+id/divideButton"
                android:text="/" />

        </GridLayout>

        <Button
            android:id="@+id/clearButton"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Clear" />

        <TextView
            android:id="@+id/resultTextView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:textSize="20sp" />

        <LinearLayout
            android:id="@+id/dynamicContainer"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical" />

    </LinearLayout>
</ScrollView>
```

5. MASTER JAVA — ACTIVITY 2 / DASHBOARD

This activity demonstrates reading Intent data, reading SharedPreferences, ListView + ArrayAdapter, normal click, AlertDialog, Toast, long-click, Context Menu, Implicit Intents, ProgressBar, fragments, FragmentManager, replace, addToBackStack and PopupMenu.

```
package com.example.mastermad;

import android.app.*;
import android.content.*;
import android.net.Uri;
import android.os.Bundle;
import android.view.*;
import android.widget.*;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;
import java.util.*;

public class DashboardActivity extends Activity {

    private ListView itemListView;
    private ArrayAdapter<String> adapter;
    private ArrayList<String> items;
    private ProgressBar progressBar;
    private SharedPreferences sharedPreferences;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_dashboard);

        TextView welcomeTextView = findViewById(R.id.welcomeTextView);
        TextView detailsTextView = findViewById(R.id.detailsTextView);

        itemListView = findViewById(R.id.itemListView);
        progressBar = findViewById(R.id.progressBar);

        sharedPreferences =
                getSharedPreferences("MasterPrefs", MODE_PRIVATE);

        // Read Intent / Bundle data
        Bundle bundle = getIntent().getExtras();

        if (bundle != null) {
            String name = bundle.getString("name", "User");
            String email = bundle.getString("email", "No email");

            welcomeTextView.setText("Welcome, " + name);
            detailsTextView.setText("Email: " + email);
        }

        // Read SharedPreferences
        String savedName =
                sharedPreferences.getString("name", "User");

        if (bundle == null) {
            welcomeTextView.setText("Welcome, " + savedName);
        }

        // ProgressBar
        progressBar.setProgress(70);

        // ListView + ArrayAdapter
        items = new ArrayList<>();
        items.add("Android");
        items.add("Java");
        items.add("XML");
        items.add("Database");
        items.add("Mobile Application");

        adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                items
        );

        itemListView.setAdapter(adapter);

        // Normal click
        itemListView.setOnItemClickListener(
                (parent, view, position, id) -> {

                    String selectedItem = items.get(position);

                    new AlertDialog.Builder(this)
                            .setTitle("Selected Item")
                            .setMessage(selectedItem)
```

```
                        .setPositiveButton("OK", null)
                        .show();

                Toast.makeText(
                        this,
                        "Clicked: " + selectedItem,
                        Toast.LENGTH_SHORT
                ).show();
            }
    );

    // Long click
    itemListView.setOnItemLongClickListener(
            (parent, view, position, id) -> {

                selectedPosition = position;
                itemListView.showContextMenu();

                return true;
            }
    );

    registerForContextMenu(itemListView);

    findViewById(R.id.homeFragmentButton)
            .setOnClickListener(v ->
                    openFragment(new HomeFragment(), false)
            );

    findViewById(R.id.detailsFragmentButton)
            .setOnClickListener(v -> {

                DetailsFragment fragment =
                        new DetailsFragment();

                Bundle fragmentBundle = new Bundle();
                fragmentBundle.putString(
                        "message",
                        "Data sent from Activity"
                );

                fragment.setArguments(fragmentBundle);

                openFragment(fragment, true);
            });

    findViewById(R.id.popupButton)
            .setOnClickListener(v -> showPopupMenu(v));
}

private int selectedPosition = -1;

// ---------------- CONTEXT MENU ----------------

@Override
public void onCreateContextMenu(
        ContextMenu menu,
        View v,
        ContextMenu.ContextMenuInfo menuInfo) {

    super.onCreateContextMenu(menu, v, menuInfo);

    menu.setHeaderTitle("Choose Action");

    menu.add(0, 1, 0, "Delete");
    menu.add(0, 2, 0, "Open Website");
    menu.add(0, 3, 0, "Call");
    menu.add(0, 4, 0, "SMS");
    menu.add(0, 5, 0, "Share");
}

@Override
public boolean onContextItemSelected(MenuItem item) {

    if (selectedPosition < 0 ||
            selectedPosition >= items.size()) {
        return super.onContextItemSelected(item);
    }

    String selectedItem = items.get(selectedPosition);

    switch (item.getItemId()) {

        case 1:
            items.remove(selectedPosition);
            adapter.notifyDataSetChanged();

            Toast.makeText(
                    this,
                    "Deleted",
                    Toast.LENGTH_SHORT
```

```
            ).show();

            selectedPosition = -1;
            return true;

        case 2:
            openBrowser("https://www.google.com");
            return true;

        case 3:
            openDialer("tel:9876543210");
            return true;

        case 4:
            openSms("sms:9876543210");
            return true;

        case 5:
            shareText(selectedItem);
            return true;
    }

    return super.onContextItemSelected(item);
}

// ---------------- IMPLICIT INTENTS ----------------

private void openBrowser(String url) {
    Intent intent = new Intent(
            Intent.ACTION_VIEW,
            Uri.parse(url)
    );

    if (intent.resolveActivity(getPackageManager()) != null) {
        startActivity(intent);
    }
}

private void openDialer(String phoneNumber) {
    Intent intent = new Intent(
            Intent.ACTION_DIAL,
            Uri.parse(phoneNumber)
    );

    if (intent.resolveActivity(getPackageManager()) != null) {
        startActivity(intent);
    }
}

private void openSms(String phoneNumber) {
    Intent intent = new Intent(
            Intent.ACTION_SENDTO,
            Uri.parse(phoneNumber)
    );

    if (intent.resolveActivity(getPackageManager()) != null) {
        startActivity(intent);
    }
}

private void sendEmail() {
    Intent intent = new Intent(Intent.ACTION_SENDTO);
    intent.setData(Uri.parse(
            "mailto:example@gmail.com"
    ));

    intent.putExtra(
            Intent.EXTRA_SUBJECT,
            "MAD Exam"
    );

    if (intent.resolveActivity(getPackageManager()) != null) {
        startActivity(intent);
    }
}

private void openMap() {
    Uri location = Uri.parse(
            "geo:0,0?q=Thiruvananthapuram"
    );

    Intent intent = new Intent(
            Intent.ACTION_VIEW,
            location
    );

    if (intent.resolveActivity(getPackageManager()) != null) {
        startActivity(intent);
    }
}
```

```
private void openCalendar() {
    Intent intent = new Intent(
            Intent.ACTION_INSERT
    );

    intent.setData(
            Uri.parse("content://com.android.calendar/events")
    );

    if (intent.resolveActivity(getPackageManager()) != null) {
        startActivity(intent);
    }
}

private void shareText(String text) {
    Intent intent = new Intent(Intent.ACTION_SEND);

    intent.setType("text/plain");
    intent.putExtra(Intent.EXTRA_TEXT, text);

    if (intent.resolveActivity(getPackageManager()) != null) {
        startActivity(Intent.createChooser(
                intent,
                "Share using"
        ));
    }
}

// ---------------- FRAGMENTS ----------------

private void openFragment(
        Fragment fragment,
        boolean addToBackStack) {

    FragmentManager manager =
            getFragmentManager();

    FragmentTransaction transaction =
            manager.beginTransaction();

    transaction.replace(
            R.id.fragmentContainer,
            fragment
    );

    if (addToBackStack) {
        transaction.addToBackStack(null);
    }

    transaction.commit();
}

// ---------------- POPUP MENU ----------------

private void showPopupMenu(View anchor) {

    PopupMenu popupMenu =
            new PopupMenu(this, anchor);

    popupMenu.getMenu().add("Edit");
    popupMenu.getMenu().add("Delete");
    popupMenu.getMenu().add("Share");

    popupMenu.setOnMenuItemClickListener(
            item -> {

                Toast.makeText(
                        this,
                        item.getTitle() + " selected",
                        Toast.LENGTH_SHORT
                ).show();

                return true;
            }
    );

    popupMenu.show();
}

// ---------------- OPTIONS MENU ----------------

@Override
public boolean onCreateOptionsMenu(Menu menu) {

    getMenuInflater().inflate(
            R.menu.menu_dashboard,
            menu
    );

    return true;
}
```

```
    @Override
    public boolean onOptionsItemSelected(
            MenuItem item) {

        if (item.getItemId() == R.id.menu_settings) {
            Toast.makeText(
                    this,
                    "Settings",
                    Toast.LENGTH_SHORT
            ).show();

            return true;
        }

        return super.onOptionsItemSelected(item);
    }
}
```

Important implementation note:

If using the platform android.app.Fragment classes above, the Fragment types and FragmentManager imports must remain from android.app. If using AndroidX instead, change the imports consistently to androidx.fragment.app.* and use androidx.appcompat.app.AppCompatActivity. Do not mix the two Fragment APIs.

6. DASHBOARD XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/welcomeTextView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Welcome"
        android:textSize="24sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/detailsTextView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <ProgressBar
        android:id="@+id/progressBar"
        style="?android:attr/progressBarStyleHorizontal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:max="100"
        android:progress="0" />

    <ListView
        android:id="@+id/itemListView"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <Button
            android:id="@+id/homeFragmentButton"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="Home" />

        <Button
            android:id="@+id/detailsFragmentButton"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="Details" />

        <Button
            android:id="@+id/popupButton"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="Popup" />
    </LinearLayout>

    <FrameLayout
        android:id="@+id/fragmentContainer"
        android:layout_width="match_parent"
        android:layout_height="180dp" />

</LinearLayout>
```

7. HOME FRAGMENT

```
package com.example.mastermad;

import android.app.Fragment;
import android.os.Bundle;
import android.view.*;
import android.widget.*;

public class HomeFragment extends Fragment {

    @Override
    public View onCreateView(
            LayoutInflater inflater,
            ViewGroup container,
            Bundle savedInstanceState) {

        return inflater.inflate(
```

```
                R.layout.fragment_home,
                container,
                false
        );
    }
}
```

fragment_home.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Home Fragment"
        android:textSize="20sp"
        android:textStyle="bold" />

</LinearLayout>
```

8. DETAILS FRAGMENT — setArguments / getArguments

```
package com.example.mastermad;

import android.app.Fragment;
import android.os.Bundle;
import android.view.*;
import android.widget.TextView;

public class DetailsFragment extends Fragment {

    @Override
    public View onCreateView(
            LayoutInflater inflater,
            ViewGroup container,
            Bundle savedInstanceState) {

        View view = inflater.inflate(
                R.layout.fragment_details,
                container,
                false
        );

        TextView messageTextView =
                view.findViewById(R.id.messageTextView);

        Bundle bundle = getArguments();

        if (bundle != null) {
            String message =
                    bundle.getString(
                            "message",
                            "No data"
                    );

            messageTextView.setText(message);
        }

        return view;
    }
}
```

fragment_details.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/messageTextView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp" />

</LinearLayout>
```

9. CUSTOM DIALOG XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/itemNameEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Item name"
        android:inputType="text" />

    <EditText
        android:id="@+id/quantityEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Quantity"
        android:inputType="number" />

</LinearLayout>
```

10. OPTIONS MENU XML

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/menu_save"
        android:title="Save" />

    <item
        android:id="@+id/menu_about"
        android:title="About" />

</menu>
```

11. DASHBOARD MENU XML

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/menu_settings"
        android:title="Settings" />

</menu>
```

12. MANIFEST

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:theme="@style/Theme.AppCompat.Light.NoActionBar"
        android:label="Master MAD">

        <activity
            android:name=".DashboardActivity"
            android:exported="false" />

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>

</manifest>
```

13. THE 10 CORE SNIPPETS TO MEMORIZE

These are the small patterns you should be able to reproduce even if the question changes completely.

1. findViewById

```
EditText nameEditText = findViewById(R.id.nameEditText);
```

2. Read text

```
String name = nameEditText.getText().toString().trim();
```

3. Button click

```
button.setOnClickListener(v -> {
    // code
});
```

4. Validation

```
if (name.isEmpty()) {
    nameEditText.setError("Enter name");
    nameEditText.requestFocus();
    return;
}
```

5. Toast

```
Toast.makeText(this, "Saved", Toast.LENGTH_SHORT).show();
```

6. Explicit Intent + Bundle

```
Intent intent = new Intent(MainActivity.this, DashboardActivity.class);

Bundle bundle = new Bundle();
bundle.putString("name", name);
intent.putExtras(bundle);

startActivity(intent);
```

7. SharedPreferences

```
SharedPreferences preferences =
        getSharedPreferences("MyPrefs", MODE_PRIVATE);

preferences.edit()
        .putString("name", name)
        .apply();

String savedName = preferences.getString("name", "");
```

8. ListView + ArrayAdapter

```
ArrayAdapter<String> adapter = new ArrayAdapter<>(
        this,
        android.R.layout.simple_list_item_1,
        items
);
listView.setAdapter(adapter);
```

9. Implicit Intent + resolveActivity

```
Intent intent = new Intent(Intent.ACTION_VIEW,
        Uri.parse("https://www.google.com"));

if (intent.resolveActivity(getPackageManager()) != null) {
    startActivity(intent);
}
```

10. Fragment data transfer

```
Bundle bundle = new Bundle();
bundle.putString("message", "Hello");
fragment.setArguments(bundle);

// inside Fragment
Bundle data = getArguments();
String message = data.getString("message");
```

14. IMPLICIT INTENT ACTIONS — EXAM TABLE

Purpose	Action / Pattern	Example
Browser / Website	ACTION_VIEW	Uri.parse("https://www.google.com")
Dialer	ACTION_DIAL	Uri.parse("tel:9876543210")
SMS	ACTION_SENDTO	Uri.parse("sms:9876543210")
Email	ACTION_SENDTO	Uri.parse("mailto:example@gmail.com")
Maps	ACTION_VIEW	Uri.parse("geo:0,0?q=Kochi")
Calendar	ACTION_INSERT	content://com.android.calendar/events
Share text	ACTION_SEND	putExtra(Intent.EXTRA_TEXT, text)

Always use resolveActivity() before startActivity() when the question expects safe handling.

15. EXAM REUSING RULE

Do not memorize every question as a completely different program. Memorize the master pattern and replace only the question-specific parts.

Master part	Change according to question
EditText fields	Name, email, mobile, patient name, product name, username, etc.
RadioButtons	Gender, payment method, room type, category, etc.
CheckBoxes	Hobbies, facilities, features, skills, add-ons, etc.
Spinner array	Course, state, country, category, vehicle, room, etc.
ListView array	Products, courses, tasks, contacts, bookings, books, etc.
Messages	Validation errors, Toast messages, dialog titles and confirmations.
Intent destination	Website, dialer, SMS, email, map, calendar or another Activity.
SharedPreferences keys	Use meaningful keys matching the question.

16. FINAL MEMORIZATION CHAIN

```
Activity 1
   ↓
findViewById()
   ↓
Read values with getText().toString().trim()
   ↓
Validate with isEmpty(), length, setError()
   ↓
RadioButton / CheckBox / Spinner / AutoComplete / RatingBar
   ↓
TextWatcher / IME Done / DatePicker / TimePicker
   ↓
SharedPreferences save/read
   ↓
AlertDialog / Custom Dialog
   ↓
Bundle + Explicit Intent
   ↓
Activity 2
   ↓
Read getIntent().getExtras()
   ↓
Read SharedPreferences
   ↓
ListView + ArrayAdapter
   ↓
Normal click → AlertDialog + Toast
   ↓
Long press → Context Menu
   ↓
Delete → notifyDataSetChanged()
   ↓
Implicit Intent
```

```
    ├── Browser
    ├── Dialer
    ├── SMS
    ├── Email
    ├── Maps
    ├── Calendar
    └── Share
    ↓
Fragments
    ↓
FragmentManager → replace() → addToBackStack()
    ↓
setArguments() → getArguments()
    ↓
Dynamic UI → addView()
```

17. IMPORTANT CORRECTION BEFORE EXAM USE

The master pattern is intentionally comprehensive, but when writing an individual lab answer, do not dump every feature into the program unless the question asks for it. Use the smallest correct subset that satisfies the question, while reusing the same syntax patterns.

Also remember: if your project uses AndroidX fragments, keep the entire Fragment implementation consistently on AndroidX. If your lab environment uses the platform Fragment API, keep it consistently on android.app.Fragment. Mixing the two APIs causes compile errors.

Exam strategy: First write the XML IDs, then Java findViewById(), then listeners, validation, data transfer and the required feature. Finally check imports and manifest entries.